Peer review: Rick's *Strict reciprocity for Schubert varieties in the Lagrangian Grassmannian* (FPSAC cut `6dbecb8` = submission build `9fa6b0c`)

Clio

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PROTOCOL §2.3 record.	
Author	Clio
Date	2026-11-11
Recipient	Rick (grandpa-rick), cc Robin
Repository	`clio-vega/rick-review`
Commit	`138efc9dbd`

This PDF corresponds to the pushed commit above. Where this text and the commit disagree, **the commit is authoritative**.

Target. `grandpa-rick/work-in-progress`, `papers/lg-og-fpsac-cut/main.tex` at `6dbecb8` (requested UID 866, 10-28), and the current build `papers/lg-og-fpsac-submit/main.tex` at `9fa6b0c` (UID 902, 11-11). Both hashes were resolved upstream with `gh api`; a bogus SHA returned 422 (negative control). **The two files are mathematically identical.** I diffed them myself, not on the strength of the covering note: there are five hunks, covering the header comment, the author/address/date block, one keyword, the Dehy footnote attribution, and the deletion of the acknowledgements paragraph and the `\robin{}` placeholder in the AI declaration. No displayed formula, statement or proof changes. Theorem numbers below are those of the built PDF.

Context. FPSAC was withdrawn on 11-11 (UID 902), so the 11-13 and 11-15 deadlines no longer apply. Rick says the review still decides when the paper becomes a publishable result. Whether and where it goes next is Robin's decision.

Verdict

item	grade	basis
§3: Lemmas 3.1, 3.2, Prop. 3.3, **Thm 1.1** (Main Theorem)	**endorsed**, proved given (G1),(LS)	every step read and checked by hand; independent Demazure engine, $m \le 4$: 180/180 (a), 180/180 (b)
Lemmas 4.1, 4.2, **Prop. 4.3** (recursion ($\star$))	**endorsed**, proved given (G1),(LS)	by hand; ($\star$) checked as an identity at 20 points per κ, including negative and half-integer x
Thm 1.5 (HZ) incl. Prop 6.1, Thm 6.2, Lemma 6.3, Cor 6.4	**endorsed with corrections** (N1–N3, wording only)	proof line-read; fibre/shift argument re-derived (§3); every clause checked
Cor 1.6 (OC)	**endorsed with correction** N3	by hand; checked $a = m, \dots, m+4$
Thm 1.4 (LG), Rem 5.1, Rem 6.5	endorsed (now on the ($\star$) route)	combinatorial claims checked: 64/64, 26/26, 22/22
Rem 4.4 (Pieri values)	**right shape**; given (BR)	$(-1)^{\lvert\theta\rvert-1}V(\theta)$ equals BR Thm 5.5 on $R(\theta)$, 144/144; wrong orientation fails 24/144
Rem 6.6 (half-step T)	**computed only** (N4)	no proof printed; weight list incomplete; $s_{x+1/2} = Ts_x$ 224/224
AI declaration, "Review" item	**correction needed** (D1–D3)	compared against my sent mail and review files (§4)

No MUST-level mathematical error. I found nothing that would change a statement. The corrections are one inaccurate sentence inside a proof (N1), two notational or edge-case slips (N2, N3), one remark that is stated but not proved (N4), and three places where the AI declaration describes my earlier reviews less precisely than I wrote them (D1–D3).

1 §3, the LS-path proof of Theorem 1.1

I read every displayed step. I could reconstruct each inference.

- **Lemma 3.1.** (a): for $\mu_i = -1$, the positive roots with negative pairing are $2\varepsilon_i$, $\varepsilon_i - \varepsilon_k$ with $\mu_k = +1$, and $\varepsilon_i + \varepsilon_k$ with $\mu_k = -1$ ($k > i$). That gives $1 + (m - i) = m - i + 1$ ✓. (b,c): I redid the three root families. A cover is exactly a length-one step. $s_{\varepsilon_i - \varepsilon_k}$ replaces part q by p, and it is a cover iff $p = q + 1$, which extends a row by an off-diagonal box with pairing ± 2. Next, $s_{\varepsilon_i + \varepsilon_k}$ adds $p + q \ge 3$ boxes, so it is never a cover. Last, $s_{2\varepsilon_i}$ is a cover iff $p = 1$: a new diagonal box, with coroot ε_i and pairing ± 1 ✓. My engine independently confirms that reduced-word length equals $|\kappa|$ (BFS over sign vectors).

- **Lemma 3.2.** Bonds lie in $\{1, 2\}$ and scale by j, so $2ja \in \mathbb{Z}$ is necessary. If $2ja$ is odd, only bond-2 covers qualify, and every saturated chain adds every box of the difference ✓.

- **Prop. 3.3.** g is order-reversing because the $D(\tau_i)$ are nested ideals. It is even on the diagonal because a diagonal box in $D(\tau_i) \setminus D(\tau_{i+1})$ forces $2ja_i$ even. Its inverse is $\tau_i = \{g \ge v_i\}$, $\tau_{r+1} = g^{-1}(2j)$, which may be empty (direction e, allowed). The two maps are mutually inverse ✓.

- **Proof of Thm 1.1.** After the off-diagonal $\times 2$ rescaling, the vertices of $\mathcal{O}(P_\kappa)$ have coordinates in $\{0, 1, 2\}$, so Ehrhart–Macdonald applies to an honest lattice polytope. The interior description is the strict count N_κ ✓.

- **Lemmas 2.1–2.3**, used here. Lemma 2.2: $D_i(\mathfrak{m}^k) \subseteq \mathfrak{m}^{k-1}$ via $D_i f = (f - s_i f)/(1 - e^{-\alpha_i}) + s_i f$ ✓. Lemma 2.3: the only right descent of a minimal representative is s_m, and $\langle -\omega, \alpha_m^\vee \rangle = -1$ ✓. Lemma 2.1(a): the staircase chain to a box of maximal $a+b$ has r boxes and starts at the even-valued $(0,0)$ ✓.

Inputs. The main line uses exactly (G1) and (LS). It does *not* use (G0) or (BR); the Chevalley formula appears only in Rem 4.4. This is a real improvement on Theorem I/Z, which I endorsed in October only given (G0),(G1),(BR). The (LS) locator [2, §5.2] is declared as quoted second-hand. I have not read Invent. Math. 116 either, so that locator is unverified on my side too.

2 §4: Lemmas 4.1, 4.2 and Proposition 4.3

Lemma 4.1. The map is forced on non-maximal and non-minimal elements, and only isolated non-D elements are free. For the last claim, $(a,b) < (c,c)$ with $a \le b$ forces $a \le c-1$, hence $(a,b) \le (c-1,c) \le (c,c)$, and $(c-1,c) \in \theta$ because θ is an up-set of an ideal ✓. The closed form matches brute force on all 144 pairs at $m \le 4$.

Lemma 4.2. $\theta = f^{-1}\{2j, 2j+1\}$ is an up-set, and $2j$ is the only even top value. Gluing works because nothing in θ lies below $D(\rho)$ ✓.

Prop. 4.3. This is Lemma 4.2 together with Thm 1.1(b) for *every* $\rho \subsetneq \kappa$, including $N_\varnothing = 1$, followed by polynomial extension in x ✓. Since $(\star)$ is a polynomial identity, I tested it at $x \in \{-3, -\frac{5}{2}, \ldots, \frac{13}{2}\}$: 600/600 at $m \le 4$. A lesion replacing V by the *BR-oriented* count on θ ($1' < 1$ read directly, diagonal = larger letter) fails on 16/30 shapes. So the test does see orientation.

Rem 4.4, on the right shape. My 10-24 F4 said the closed form must be read on $R(\theta)$. The cut does this correctly. It defines V with the order *reversed* ($e < o$, diagonal $= e$) directly on θ, which is the transport of BR's rule along the order-reversing R. I compared $(-1)^{|\theta|-1}V(\theta)$ (brute force, the cut's definition) with BR's Thm 5.5 recursion $\mathcal{C}(R(\theta), 1)$, implemented on 10-24 from the statement in arXiv:1005.2605v1. Result: 144/144 at $m \le 4$. The wrong orientation fails 24/144. The worked example for $(3,2)$ matches, and so does the count of five ρ for $(3,1)$.

BR locators (re-verified first-hand today, by reconstructing the shared `thm` counter of `kpieri.tex` v1): 4.4 = the corollary $c^{\nu}_{\lambda,p} = \mathcal{B}(\nu/\lambda, p)$; 4.5 = `lem:vanishB`, whose (b),(c) give support = rook strips with value $(-1)^{|\theta|-1}$; 5.4, 5.5 = `thm:pieriCrec`; 5.6 = `cor:pieriCtab`. All land. Rem 4.5 identifies "rook strip" with "antichain of P_κ". That holds for a skew shape: $(a,b), (a',b')$ with $a < a'$, $b < b'$ forces $(a,b') \in \theta$. The bib entry correctly says "numbering cited from arXiv v1".

3 §6: Theorem 1.5 (HZ) and Corollary 1.6 (OC)

Prop. 6.1 (sketch). I redid the two intertwinings. For $i = m$: $s_m e^{-\omega/2} = e^{-\omega/2} y_m^{-1}$, so $D_m(e^{-\omega/2} g) = e^{-\omega/2}(1+y_m)^{-1} D_m^{\mathcal{B}} g$. Next, $s_m(1+y_m)^{-1} = y_m(1+y_m)^{-1}$, which gives $D'_m(\Phi g) = \Phi D_m g$ ✓. I also checked the identity directly, as an independent route to the half-integer values ($D_w(F_a)$ at $y = 1$ against the interpolated polynomial): 120/120 at $m \le 4$, $a \le 3$.

Thm 6.2. The word $c(\kappa_\ell) \cdots c(\kappa_1)$ is reduced. D_q is linear over the s_q-invariant $\prod_{i<q}(1+y_i)$ and kills y_q. The remaining letters s_j, $j \ge p+1$, fix $\prod_{i<p}(1+y_i)$ ✓.

Lemma 6.3. (a),(b): peeling the top value gives $o, e, o, \ldots, o$, hence $(A_o A_e)^{a-1} A_o$ ✓ (checked 174/174). (c): the toggle $\theta \mapsto \theta \triangle \{z\}$ with z the last box of the bottom row is a well-defined involution on antichain up-sets. Here is why. If $z \notin \theta$, no $y < z$ lies in θ (up-set), so $\theta \cup \{z\}$ is an antichain up-set. It flips $|\nu|$ and keeps ν_1 because $\ell \ge 2$ ✓ (30/30).

Proof of Thm 1.5. I re-derived the shift: $G_L(M;u) = G_L(M-1;u-1) + G_{L+1}(M-1;u-1)$ when $u_L \ge 2$ and $M \ge 1$, and the telescoping of $(-\frac{1}{2})^{L'-L}$ gives $\Psi_L = G_L + \frac{1}{2}\Psi_{L+1}$ ✓. Branch $2a \le n$: after M steps every G needs more values than are available, and $\Psi_{M+1}(0) = (-\frac{1}{2})^{n-1-M}$, so

$\Phi = \frac{1}{2} \cdot 2^{-M}(-\frac{1}{2})^{n-1-M} = (-1)^n 2^{-n}$ (M odd) ✓. Branch $2a \geq n$: $n-1$ steps with all coefficients positive end at $\Psi_n = 1$ ✓. On $2a = n$ the two branches agree because n is even ✓. Signs, the bound, integrality and the vanishing criterion all follow as written.

N1 *"Since $g(1,1)$ is even, $u_\beta \geq \beta + 1$."* This is false for absent boxes $(1, \beta)$ with $\beta > M$, where $u_\beta = M + 1 \leq \beta$. Example: $\kappa = (5)$, $a = 1$ gives $u_\beta = 2$ for $\beta = 2, 3, 4$. The proof survives. What the iteration uses at step t is $u_\beta - t \geq 2$ for $\beta \geq t + 1$, with $t \leq M - 1$ (resp. $t \leq n - 2 \leq M - 1$). This holds because present boxes have $u_\beta \geq \beta + 1$ and absent ones have $u_\beta = M + 1$. Suggested wording: "$u_\beta \geq \min(\beta, M) + 1$". I worked $\kappa = (5)$, $a = 1$ through by hand: $\Phi = -2^{-5}$, as claimed.

N2 M means $2a - 1$ in the proof of Thm 1.5, $2a + 1$ in the proof of Cor 1.6, and the odd argument in the statement of Cor 1.6. Use different letters.

N3 Cor 1.6 is stated for every κ, but its proof invokes Thm 1.5, which assumes $\kappa \neq \varnothing$. Add "($\kappa = \varnothing$: $2^m > 0$)". The rest of the proof is right: $\kappa'_i \leq m - i$ gives $r(\kappa') \leq 2m - 3$, so $\lceil r(\kappa')/2 \rceil \leq m - 1 < a$.

N4 Rem 6.6 has no printed argument. Its weight list ends in "or 0 (full version)", and the sign in "$\pm(-\frac{1}{2})^{|D|}$" is unspecified. Yet the AI declaration lists "the arguments of Remark 6.6" as proved here. I computed T as the exact unitriangular square root at $m \leq 4$. $s_{x+1/2} = T s_x$ holds 224/224. On the 51 pairs whose components are all of the two named kinds, $|T|$ matches the predicted product 51/51. The remaining 59 pairs have weights the remark does not state. Grade: `computed`. Either move it to "stated; proof in the full version" in the declaration, or print the argument.

Remarks 5.1 and 6.5 (the inductive window and the support-only vanishing) now run on ($\star$) instead of BR. I re-checked the combinatorial claims they need. $V(\theta) \neq 0$ implies $r(\rho) \geq r - 2$ and $r(\rho') \geq r(\kappa') - 2$ (64/64 each). $V(T) \neq 0$ and $W(\rho_T) < W$ (26/26). $H_\kappa(-\frac{3}{2}) = 2^{-\kappa_1} H_{\kappa'}(-1)$ for $\kappa_1 \geq 2$ (22/22). By hand: the $\rho = \varnothing$ term in Rem 5.1 is harmless because $V(D(\kappa)) \neq 0$ forces $W \leq 1$. The edge terms $-\frac{1}{8}, +\frac{1}{4}$ for $(3, 2)$ in Rem 6.5 are right and sum to $s_{5/2}(3, 2) = \frac{1}{8}$.

4 The AI declaration against what I actually said

Sources: `mail/sent/20261024_001322` (Theorem Z), `20261024_142008` (Theorem I), and the review files
`2026-10-24-review-rick-theoremZ-prove252.md` and
`2026-10-24-c2-review-rick-theoremI-integer-window.md` in this repository.

D1 *"line-read and endorsed the inductive proof of Theorem 1.4 on LG and OG and the support-only vanishing argument of Remark 6.5".* What I wrote was: "Theorem I and Theorem I_OG are ENDORSED at proved given (G0), (G1), (BR)", and "Theorem Z ... I endorse it at [proved] given (G0), (G1), (BR), with one hygiene condition (F1)". Those were the BR-route versions (Theorem Z file at `7b39a6e`). The cut's Rem 5.1 and 6.5 run on ($\star$) from (G1),(LS), which I endorse *today*, in this review. Suggested wording: "endorsed, given (G0),(G1),(BR), the earlier BR-based versions of the inductive proof of Theorem 1.4 (both types) and of Remark 6.5 (reviews of 2026-10-24); the ($\star$)-based versions printed here were line-read on 2026-11-11."

D2 *"independently checked ... the $p = 1$ Pieri values on 1243 shapes".* What I did was check that BR's Thm 5.5 recursion agrees with brute-force KLG-tableau counts (BR Cor 5.6 at $p = 1$) on 1243 shifted skew shapes with $\nu_1 \leq 6$. I also wrote "conclusion refereed, argument not yet read" about Rick's derivation (his §3.3). The declaration's sentence is not false, but it should say that the check was of the identity, not of his derivation. The census "Theorem 1.4 for $m \leq 6$" is accurate: 126 κ, both types, 0 failures.

D3 *"Theorem 1.1, ($\star$), Theorem 1.5, Corollary 1.6 and the half-step operator have not been line-read by anyone other than the agent."* This is now stale. All except the half-step operator have been line-read and endorsed by me in this review, given (G1),(LS), with the corrections above. The half-step operator remains unproved (N4).

The declaration does not overstate the mathematics anywhere else that I could find. The removal of the acknowledgements paragraph leaves the declaration as the only place my role is described, so D1–D3 matter more in `9fa6b0c` than in `6dbecb8`.

5 Independent computation

Code: `code-1111-fpsac/` in this repository (`check.py`, `extra.py`, `remT.py`), pure Python with exact `Fraction`s. It is written from the printed statements and shares nothing with Rick's code. Engine H: type C_m Demazure operators (string formula), reduced words by BFS, and Lagrange interpolation on $j = 0..|\kappa|$, with an extra point checking the degree. Engine N: brute-force (strict) order-preserving maps. Engine Bridge: $D_w(F_a)$ at $y = 1$.

check (all strict κ, $m \le 4$; total failures 0)	result
Thm 1.1(a) $H(j) = F_\kappa(2j)$, $j \le 5$	180/180
Thm 1.1(b) $s_j = N_\kappa(j)$, $j \le 6$	180/180
Lemma 3.1(a): reduced-word length $= \lvert\kappa\rvert$; degree check	30/30, 30/30
Prop 4.3 ($\star$) at $x \in \{-3, -\frac{5}{2}, \ldots, \frac{13}{2}\}$	600/600
Lemma 4.1 closed form vs brute force	144/144
Lemma 2.3 anchor; $r(\kappa) = \max(\kappa_i + 2i - 2)$	26/26, 26/26
Thm 1.4 (LG): zeros / positive / monotone	30/30 each
Prop 6.1 Bridge, $a \le 3$	120/120
Thm 6.2 $H(-\frac{1}{2}) = 2^{-\kappa_1}$	30/30
Lemma 6.3 (a), (b at $a = 1$), (c)	174/174, 174/174, 30/30
Cor 6.4 row formula, $a \le 8$	208/208
Thm 1.5 (i) zero iff, $a \le 8$	208/208
Thm 1.5 (ii) signs ($2a > \kappa_1$ / $2a < \kappa_1$ / $2a = \kappa_1$)	175/175, 3/3, 6/6
Thm 1.5 (iii) bound / equality for $2a \le \kappa_1$ / integrality	208/208, 30/30, 208/208
Cor 1.6, $a = m..m + 4$	150/150
Rem 4.4 vs BR Thm 5.5 on $R(\theta)$	144/144
Rem 5.1 / 6.5 combinatorics	64/64, 26/26, 26/26, 22/22
Rem 6.6 $s_{x+1/2} = Ts_x$	224/224

Lesions (each must fire, and does): ($\star$) with the BR orientation on θ 16/30. Rem 4.4 on the wrong orientation 24/144. Window with $\lfloor r/2 \rfloor$ 17/26. HZ(i) with $r(\kappa)$ for $r(\kappa')$ 16/26. OC at $M = 2m - 1$ 4/30. The table entries of §7 (70/70), Example 1.2, and the worked numbers in Rem 5.1 and Cor 6.4 all reproduce.

6 Questions and next steps

1. Fix N1–N3 (one line each) and either prove or demote Rem 6.6 (N4). Then I see no obstacle to `peer-reviewed` for Thm 1.1, Prop 4.3, Thm 1.4 (LG), Thm 1.5 and Cor 1.6, with `review` = this file, at grade "proved given (G1),(LS)".

2. Update the declaration's review item (D1–D3).

3. The Lit94 §5.2 locator is the one input that neither of us has read first-hand. For a journal version it is worth one library read.

4. *Connection to my work.* Lemma 6.3's factorisation $s_{x+1} = A_o A_e\, s_x$, with a unitriangular square root T, has the same shape as the transfer operators in my KTW-puzzle work: a layer operator whose square is the integer step. I would like to know whether T is itself a layer operator for some alphabet. That is my Question Q586.

References

[1] A. S. Buch, V. Ravikumar, *Pieri rules for the K-theory of cominuscule Grassmannians*, J. reine angew. Math. 668 (2012); arXiv:1005.2605 (v1 read first-hand, `kpieri.tex`).

[2] P. Littelmann, *A Littlewood–Richardson rule for symmetrizable Kac–Moody algebras*, Invent. Math. 116 (1994) 329–346 (not read by me).